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Bridging Viewpoint Gaps: Geometric Reasoning Boosts Semantic Correspondence

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Abstract

Finding semantic correspondences between images is a challenging problem in computer vision, particularly under significant viewpoint changes. Previous methods rely on semantic features from pre-trained 2D models like Stable Diffusion and DINOv2, which often struggle to extract viewpoint-invariant features. To overcome this, we propose a novel approach that integrates geometric and semantic reasoning. Unlike prior methods relying on heuristic geometric enhancements, our framework fine-tunes DUSt3R on synthetic cross-instance data to reconstruct distinct objects in an aligned 3D space. By learning to deform these objects into similar shapes using semantic supervision, we enable efficient KNN-based geometric matching, followed by sparse semantic matching within local KNN candidates. While trained on synthetic data, our method generalizes effectively to real-world images, achieving up to 7.4-point improvements in zero-shot settings on the rigid-body subset of Spair-71K and up to 19.6-point gains under extreme viewpoint variations. Additionally, it accelerates runtime by up to 40 times, demonstrating both its robustness to viewpoint changes and its efficiency for practical applications. Code is available at `https://github.com/Qiyang-Q/Bridge-Viewpoint-Gap`.

1. Introduction

Finding semantic correspondences aims at the identification of semantically similar points across different images. Establishing these correspondences enables various applications, such as content editing, visual retrieval, and style transfer, etc. The challenge of this task arises from both the variations in appearance and the changes in viewpoint. Recent vision representation models [5, 16, 42, 50] have made significant strides in learning invariant representations that are robust to changes in appearance. Yet, these models are primarily trained on 2D images and lack an understanding of 3D geometry. This limitation makes it difficult for these models to find accurate semantic correspondences when there are substantial viewpoint changes.

To tackle this issue, previous work has attempted to incorporate geometric awareness into semantic features to make representations more robust to viewpoint variations and improve 3D understanding. For example, Telling Left from Right [61] tries to incorporate perspective information by finetuning on some 2D direction basis like left and right, but this is not sufficient to handle drastic 3D perspective changes. Spherical map [34] back-projects the 2D features into a spherical space to guide the semantic correspondences. However, it requires supervised finetuning on specific target domains. To achieve generalizability, Xu *et al*. [58] try to directly inject the 3D awareness into Stable Diffusion by training a ControlNet on large-scale posed videos. However, it can only accommodate minor viewpoint alterations. Moreover, finding the best correspondences on dense feature maps is computationally intensive, thereby impeding their practical application.

To address these challenges, we are inspired by a fundamental principle: semantic correspondences exhibit geometric consistency in 3D space, regardless of viewpoint variations in 2D images. Therefore, we propose a new approach: instead of injecting geometric awareness when learning the semantic features, we bridge the viewpoint gap by first establishing coarse correspondences purely through geometric reasoning then refining the correspondences using semantic alignment. Specifically, we make a novel use of the advanced model, DUSt3R [54], a robust model originally designed for 3D reconstruction from two views.

Unlike conventional reconstruction tasks that rely on two views of the same object, we fine-tune DUSt3R with cross-instance data to reconstruct two distinct 3D objects within a unified 3D space. To accomplish this, we leverage the extensive synthetic data from ShapeNet [6], collecting cross-instance image pairs with significant viewpoint variations. After finetuning on the synthetic data, DUSt3R is capable of reconstructing two instances simultaneously with aligned poses. Afterwards, we introduce a strategy for learning 3D

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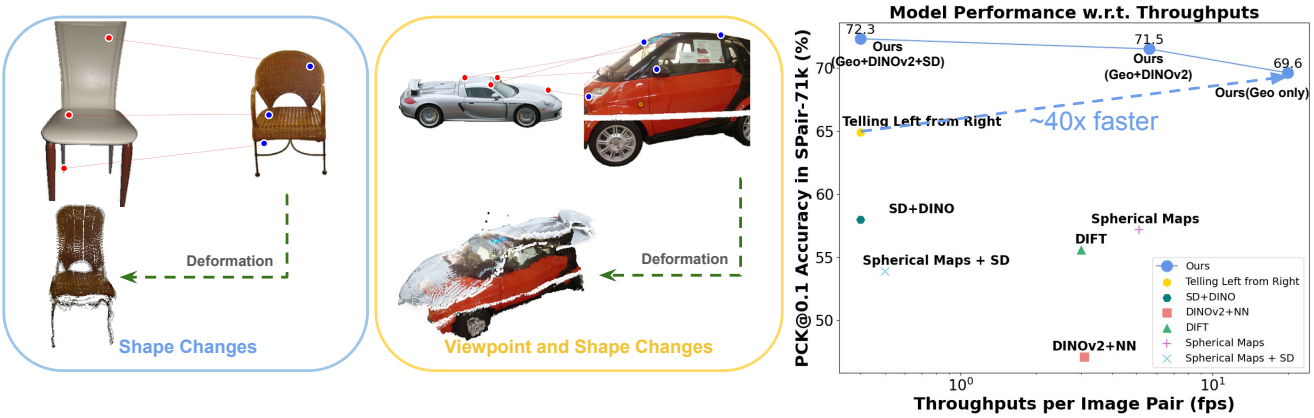


Figure 1. Our method bridges the viewpoint gaps through geometric reasoning for finding semantic correspondences. **Left image:** We deform the different instances into similar shape. **Middle image:** After deformation, we can find the correspondences even under drastic viewpoint change. **Right image:** We achieve state-of-the-art performance in semantic correspondences with high efficiency.

deformations between the pair, so that the two different objects can be mapped into similar morphologies (as shown in the left part of Fig. 1). Following deformation, we can directly apply simple K-nearest neighbour matching to the 3D objects. Note that this was previously impossible because these two objects are originally presented in different shapes and viewpoints. Following this, we apply semantic matching to refine the coarse correspondences. Different from previous methods that conduct densely feature-by-feature matching, we rely on geometric reasoning to identify semantically similar local regions and only sparsely match their semantics, which significantly enhances computational efficiency. Our method is a new pipeline that combines geometric and semantic matching.

Extensive experiments conducted on the Spair-71K dataset demonstrate both the efficacy and efficiency of our proposed approach. Our approach is evaluated under three scenarios: (1) using geometric reasoning without semantic matching, (2) incorporating semantic reasoning to refine semantic correspondences using DINO V2 features, and (3) employing both DINO V2 and Stable Diffusion features for semantic reasoning. Note that even though DUSt3R is fine-tuned with synthetic datasets, experimental data indicates that our method is capable of generalizing to real-world images, with all settings achieving significant gains over state-of-the-art methods, as shown in Fig. 1. Notably, relying solely on geometric reasoning, our method exceeds the prior art [61] by a margin of 4.7 points on the rigid-body subsets of Spair-71K dataset, while achieving approximately 40 times faster inference speed. When using DINOV2 combined with Stable Diffusion for semantic refinement, we surpass the state-of-the-art by 7.4 points. Notably, our approach demonstrates remarkable resilience to drastic viewpoint changes. It enhances the zero-shot method [61] by up to 19.6 points in scenarios with significant viewpoint variations and the in-domain fine-tuning method [33] by up to 8.7 points. These results endorse our novel pipeline that synergizes geometric and semantic reasoning. Our contributions are summarized below:

- We introduce an innovative approach for identifying semantic correspondences: unlike prior methods that enhance semantic features through geometric bootstrapping, our pipeline is grounded in geometric reasoning and leveraging semantic reasoning for refinement.
- Our method improves the state-of-the-art method by a large margin while speeding up the runtime by up to 40 times. More importantly, our method in a zero-shot manner is robust to significant changes in viewpoint even outperforming in-domain finetuned method, which highlights its vast potential for practical use.

2. Related Work

Image Correspondence. Establishing correspondences between images is a fundamental task in computer vision, with applications in Structure-from-Motion/3D reconstruction [1, 39, 45, 46], object tracking [12, 59], image recognition [4, 40, 49] and segmentation [15, 25, 28, 43]. Visual correspondence and semantic correspondence, while differing in whether they focus on the same instance or different instances within the same category, share the common goal of learning features invariant to illumination, occlusion, articulation, viewpoint, and even semantic changes. Traditional methods rely on hand-crafted features such as SIFT [30] and SURF [3] to find correspondences. With the advent of deep learning, convolutional neural networks (CNNs) have been employed to learn feature representations for correspondence tasks. Supervised methods have shown promising results by learning representations from ground-truth correspondence annotations [9, 18, 22, 23]. To address the scarcity of labeled data, weakly supervised [52] and self-supervised methods [20, 56] have been developed,

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allowing models to learn correspondences without explicit annotations. Recently, transformers have been introduced to correspondence problems. Methods using transformers leverage attention mechanisms to learn robust representations that capture both spatial relationships and visual appearance [22, 27, 44, 53].

Pre-trained Representations from Generative and Foundation Models. Recent works have utilized representations learned from generative models, such as Generative Adversarial Networks (GANs) [13], diffusion models [17, 21, 47, 48], and visual foundation models such as DINO [5, 38] and OpenCLIP [19], to improve semantic correspondence. These pretrained representations exhibit an emergent ability to capture rich semantic information originating from tasks like image generation and generalize effectively across various datasets and tasks. Such methods have changed the paradigm of finding semantic correspondences by providing features that can be used directly for matching [50, 60]. Subsequent methods have investigated how to use these features as frozen backbones and add additional modules for better correspondence [31, 60, 61]. However, these representations are not without limitations. Recent works [60, 61] point out that these representations struggle when handling large viewpoint changes, indicating that they lack 3D awareness.

Learning 3D-Aware Representations. To address the limitations of 2D representations under significant viewpoint changes, some previous work has investigated how to incorporate 3D information into representations learned from 2D image planes. Techniques like Neural Radiance Fields (NeRF) [35] learn implicit representations through volumetric rendering, and NeRF-based object detection [57] demonstrates that such representations perform well on 3D detection tasks. Other methods use camera pose or object pose information to learn invariant features with respect to changes in camera view [2, 7, 56, 58]. In addition, some approaches extend this idea to semantic correspondence. For example, [61] introduces a pose-variant augmentation in test time through flipping to align images, while [34] injects geometric priors into learned representations using viewpoint information.

Explicit 3D Reasoning. Some previous work [26, 51] attempted to directly map image pixels into a 3D space by predicting a canonical 3D coordinate for each pixel. However, these methods force the network to learn a fixed coordinate space. Recently DUSt3R [54] proposes to jointly map two images in a shared 3D coordinate space, allowing robust cross-view alignment. Based on DUSt3R, MASt3R [24] refines this idea for image correspondence, highlighting the ability of such joint 3D representations to produce accurate matches across views.

3D Deformation for dense correspondence. While mapping images from the same category into a shared 3D coordinate space can effectively capture global spatial relationships, such approaches struggle with intra-category geometric variations. Alongside direct 3D mapping, some approaches use point cloud deformation to establish dense correspondences between category-level objects. For instance, [55] learns per-vertex offsets to warp a source model, [8] maps point clouds to a canonical primitive and deforms them for correspondence, and [10] represents 3D shapes as deformed templates implicit field using dedicated deformation and correction fields. These works show that existing 3D deformation methods can capture geometric variations, enabling robust semantic correspondence.

Building upon these insights, we propose a method that combines the advantages of pre-trained representations from visual foundation models with explicit 3D geometric reasoning. By integrating semantic features extracted from models like DINO with joint 3D coordinate mapping and deformation-based refinement, our approach effectively handles semantic correspondence under large viewpoint changes.

3. Proposed Approach

3.1. Problem Setup

We aim to establish pixel-level correspondences between two images that contain the same instance or similar instances within one category. Let I^1 be the source image and I^2 be the target image, with a set of keypoint pixel locations p^1 in I^1. Our goal is to find the corresponding pixel locations p^2 in I^2 such that they not only share similar semantic meanings but also preserve the geometric consistency of the instances in I^1 and I^2.

A common paradigm in previous work for obtaining semantic correspondences is to first extract dense image features from both images, denoted as $\mathcal{F}^1$ and $\mathcal{F}^2$, and then match keypoints by minimizing the distance between feature descriptors:

$$p^2 = \arg\min_p d\left(\mathcal{F}^1(p^1), \mathcal{F}^1(p)\right) \tag{1}$$

where d is typically the cosine distance used as the similarity metric.

Although such approach can learn shape or semantic invariant representations, it struggles to capture view-invariant representations solely from the image plane. This limitation makes semantic correspondence under large viewpoint changes a challenging problem.

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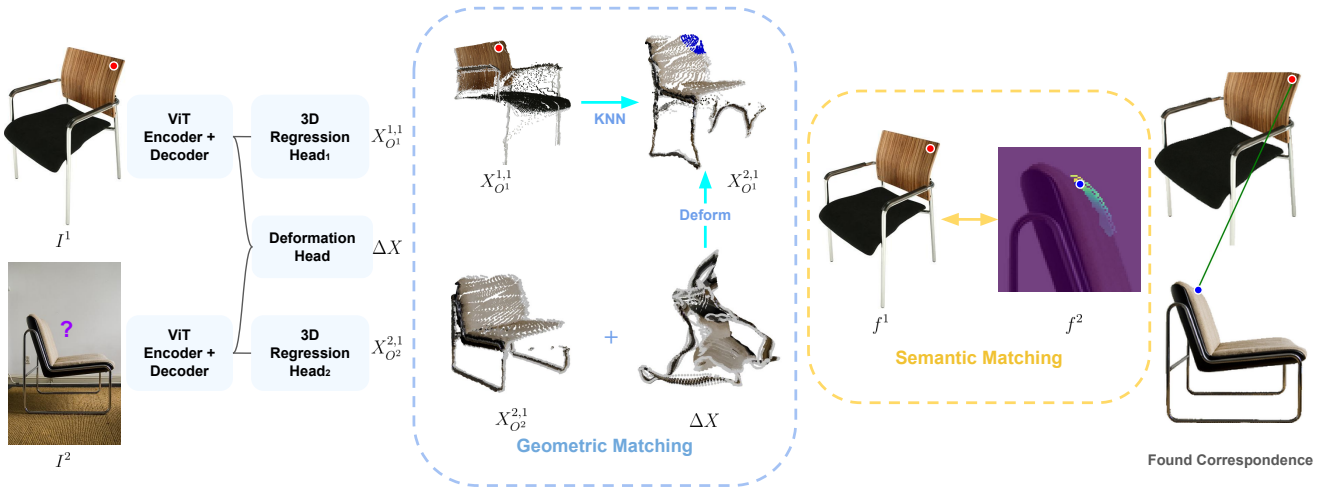


Figure 2. The framework of our method.

To address this issue, we propose to explicitly lift both images into a unified 3D space and perform matching in the 3D space rather than in feature space, as depicted in Fig. 2. Specifically, we reconstruct the shapes of objects in both I^1 and I^2 as 3D point maps within an aligned coordinate system. We then deform the shape of the object in I^2 to align with the shape of the object in I^1, while maintaining the camera pose in I^2. This process results in $\mathcal{X}^1$ and $\mathcal{X}^2$, which represent the pixelwise 3D coordinates for I^1 and I^2, respectively. Thus, we obtain semantic correspondence as follows:

$$p^2 = \arg\min_p d\left(\mathcal{X}^1(p^1), \mathcal{X}^2(p)\right), \tag{2}$$

where d is the euclidean distance.

3.2. Image to 3D Framework

Our model is built upon DUSt3R [54], which employs a transformer-based network to perform scale-free 3D reconstruction from two input RGB images, resulting in two dense 3D point clouds, or pointmaps, designated as $X^{1,1}$ and $X^{2,1}$.

A pointmap $X^{a,b} \in \mathbb{R}^{H\times W\times 3}$ represents a dense mapping of each pixel $i = (u, v)$ in the image I^a to its corresponding 3D point $X^{a,b}_{u,v} \in \mathbb{R}^3$, expressed in the coordinate system of camera $\mathcal{C}^b$. DUSt3R jointly predicts pointmaps $X^{1,1}$ and $X^{2,1}$ in the same coordinate system of camera $\mathcal{C}^1$.

Network Architecture. Both input images I^1 and I^2 are encoded using a shared-weight Vision Transformer (ViT) encoder [11], producing token representations F^1 and F^2:

$$F^1 = \text{Encoder}(I^1), \quad F^2 = \text{Encoder}(I^2). \tag{3}$$

These representations are jointly processed by a transformer decoder that exchanges the information through cross-attention for 3D reasoning across the viewpoints of I^1 and I^2, resulting in G^1 and G^2:

$$G^1, G^2 = \text{Decoder}(F^1, F^2). \tag{4}$$

Finally, 3D regression heads regress the pointmaps and confidence maps from decoder outputs:

$$X^{1,1}, Conf^1 = \text{Head}^1([F^1, G^1]), \tag{5}$$
$$X^{2,1}, Conf^2 = \text{Head}^2([F^2, G^2]). \tag{6}$$

By jointly processing both images and exchanging information via cross-attention, the network learns spatial relationships necessary for accurate 3D reconstruction and alignment, enabling reliable semantic correspondence under large viewpoint changes.

3.3. Learning Category-Level 3D Geometry

We extend the DUSt3R framework [54] to operate at the **category level**, enabling our model to understand spatial relationships between different objects within the same category from various points of view. To achieve this, we use renderings from the ShapeNet dataset [6], which provides pose-aligned categorical 3D models. Therefore, it ensures that renderings from the same camera pose across all objects in a category result in orientation-consistent images.

We denote the object in I^1 and I^2 by O^1 and O^2. Our model predicts pointmaps $X^{1,1}_{O^1}$ and $X^{2,1}_{O^2}$ corresponding to input images I^1 and I^2, respectively, expressed in the coordinate system of camera $\mathcal{C}^1$. Importantly, the pointmaps produced by our model from the 3D regression head do not enforce the shape consistency between the objects in I^1 and I^2. In other words, the two 3D regression heads produce shapes that are aligned in orientation but allow variations in object structure, capturing the unique characteristics of individual instances within the same category.

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3.4. Deformation Head for Shape Morphing

Establishing accurate correspondences between the predicted point maps $X^{1,1}_{O^1}$ and $X^{2,1}_{O^2}$ can be framed as learning a deformation field that morphs the shape of O^2 and O^1. To this end, we introduce a *deformation head* to capture the semantic and geometric mapping between O^1 and O^2.

The deformation head combines the encoder outputs F^1, F^2 and the decoder outputs G^1, G^2 from both input images as the input. This combined representation is then used to predict the 3D offsets for each pixel in I^2 and the associated confidence map:

$$\Delta X, \mathit{Conf}^{\text{def}} = \text{Head}^{\text{def}}\left([F^1, G^1, F^2, G^2]\right), \tag{7}$$

where ΔX represents the per-pixel 3D offset from the shape of O^2 to the shape of O^1. The resulting pointmap after deformation is formulated as:

$$X^{2,1}_{O^1} = X^{2,1}_{O^2} + \Delta X, \tag{8}$$

where $X^{2,1}_{O^1}$ is the deformed 3D pointmap that matches the shape of O^1 in the coordinate system of camera $\mathcal{C}^1$, while maintaining the 2D pixel correspondence to image I^2. With the correct deformation, correspondences can be established in the overlapping regions between $X^{2,1}_{O^1}$ and $X^{1,1}$. Therefore, we can substitute X^1, X^2 in Eq. (2) with $X^{1,1}, X^{2,1}_{O^1}$ to find pixel correspondences in 2D images.

3.5. Deformation Loss with Semantic Supervision.

The deformation from $X^{2,1}_{O^2}$ to $X^{2,1}_{O^1}$ is essential for establishing accurate semantic correspondences. Beyond aligning surface geometry, it must ensure semantic consistency between the parts of two objects. To this end, we propose training the deformation head using a novel semantic correspondence supervision approach, which eliminates the need for manual labeling by distilling knowledge from DINOv2 [38] in a manner that is robust to viewpoint changes.

The core idea of our label-free semantic supervision approach is to leverage RGB renderings of different objects from the same view to automatically generate pseudo-labels for semantic correspondences. Let $\mathcal{C}^2$ be the camera of interest, we denote the RGB images of the two objects as $I^2_{O^1}$ and $I^2_{O^2} = I^2$. In the absence of viewpoint variations, existing feature extractors can accurately identify these correspondences. By combining these pseudo-labels with 3D geometric correspondences derived from different views of the same object, we can obtain robust supervision signals between two images of different views and different objects.

We employ the DINOv2 feature extractor to obtain dense feature maps of I^2 and $I^2_{O^1}$, denoted as f^2 and $f^2_{O^1}$, respectively. During training, given the ground truth pointmaps X^{gt}_2 and $X^{\text{gt}}_{2_{O^1}}$ for I^2 and $I^2_{O^1}$, we proceed as follows: For each feature corresponding to a pixel in f^2, we find its K nearest neighbors in $X^{\text{gt}}_{2_{O^1}}$ within the 3D space. This ensures that cosine similarity computations are confined to a local 3D region, enhancing the relevance conditioned on spatial proximity. We compute the cosine similarity between f^2 and $f^2_{O^1}$ within these local neighborhoods. Then, we select correspondences in which the cosine similarity exceeds a certain threshold τ to serve as sparse pseudo labels for semantic correspondence supervision. Formally, the set of labeled correspondences $\mathcal{S}$ is defined as:

$$\mathcal{S} = \left\{(i,j) \;\middle|\; d\left(f^2_i, f^2_{O^1,j}\right) > \tau,\ j \in \text{NN}_K\left(X_2, X^{\text{gt}}_{O^1}, i\right)\right\} \tag{9}$$

where i indexes pixels in f^2, j indexes pixels in $f^2_{O^1}$, $\text{NN}_K(\cdot)$ denotes the set of K nearest neighbors in 3D space and the distance metric d is the cosine distance.

With the semantic supervision established, we define the semantic-aware deformation regression loss as:

$$\mathcal{L}_{\text{semantic}} = \sum_{i \in S}\left(\mathit{Conf}^{\text{def}}_i \left\|\mathbf{X}_i - \bar{\mathbf{X}}_i\right\|^2_2 - \alpha \log \mathit{Conf}^{\text{def}}_i\right) \tag{10}$$

where:

- $\mathbf{X}_i$ is the predicted deformed 3D point for the pixel i, $\mathbf{X}_i = X^{2,1}_{O^2 i} + \Delta X_i$.
- $\bar{\mathbf{X}}_i$ is the ground truth 3D point corresponding to the DINO-found correspondence for the pixel i.
- $\mathit{Conf}^{\text{def}}_i$ is the confidence score for the pixel i predicted together with ΔX.
- α is a hyperparameter that controls regularization term.

We apply a regression loss with confidence weighting, following [54], to encourage the network to prioritize well-defined correspondences. Ambiguous correspondences are assigned lower weights, partially allowing the network bias to interpolate them from more confident correspondences.

In addition to this sparse semantic supervision, we apply dense supervision on the deformed shape using the Chamfer distance loss $\mathcal{L}_{\text{Chamfer}}$, which measures the discrepancy between the predicted deformed pointmap $X^{2,1}_{O^1}$ and the ground truth pointmap $X^{2,1\text{gt}}_{O^1}$, the chamfer distance loss is formulated as:

$$\mathcal{L}_{\text{Chamfer}} = \sum_{p \in X^{2,1}_{O^1}} \min_{q \in X^{2,1\text{gt}}_{O^1}} \|p-q\|^2 + \sum_{q \in X^{2,1\text{gt}}_{O^1}} \min_{p \in X^{2,1}_{O^1}} \|q-p\|^2. \tag{11}$$

Dense supervision from chamfer distance loss encourages the overall shape after deformation to closely match the ground truth, ensuring geometric accuracy alongside semantic alignment. Therefore, combining geometric and semantic supervision, our final deformation loss is $\mathcal{L}_{\text{deform}} = \mathcal{L}_{\text{semantic}} + \mathcal{L}_{\text{chamfer}}$.

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3.6. Geometric and Semantic matching

With the deformation head, two images in different instances are now not only aligned in orientation but also matched in shape and semantics in the unified coordinate space. During inference time, we can establish correspondences by geometric matching using 3D nearest neighbor. In addition to geometric matching, we further propose a test-time semantic matching approach, formulating a hybrid matching strategy for finer-level semantic correspondence.

For coarse geometric matching, we use the predicted point maps of the source and target images, X^1 and X^2, to construct k-d trees for efficient nearest neighbor searches in 3D space. Given a set of keypoints P in the source image, we identify the K nearest neighbors in 3D space from the target image's pointmap X^2. These neighbors serve as candidate matching points in the target image.

For fine-grained semantic matching, we extract dense semantic feature maps f^1 and f^2 for both images using a pre-trained feature extractor. For each keypoint in P, we calculate the cosine similarity between its semantic feature in f^1 and the semantic features in f^2 corresponding to the K nearest neighbors identified during geometric matching. The final correspondence is determined by points with highest similarity from the candidates.

4. Experiments

4.1. Implementation details

Training Dataset. We generate a comprehensive and photorealistic dataset using ShapeNetCore[6] 3D models that provide consistently categorized pose-aligned 3D objects. For each object within the same category, we first normalize its size to ensure uniformity across all instances by scaling it using the factor $s = \frac{1}{\max(x,y,z)}$, where x, y, z are the lengths of the object's bounding box in 3 dimensions. We then rendered both RGB and depth images for each normalized object using 100 predefined camera rotations with uniform increments in azimuth and elevation angles, ensuring comprehensive coverage of views and consistency of the same camera pose in different objects across the entire category. To mitigate the simulation-to-reality gap and enhance the realism of our dataset, we utilize Blender's ray tracing engine with High Dynamic Range Imaging (HDRI) environments for real-world illumination conditions. As a result, we generate an extensive dataset, covering 55 common object categories, comprising 5,130,000 RGB and depth images at a resolution of 512×512 pixels, using 51,300 models from ShapeNetCore, resulting in a total dataset size of approximately 1.3 terabytes.

Inference in real world image. We first generate a mask of the object in the image following [61] using the SAM2 model [41]. In the geometric matching stage, we predict 3D pointmaps for both the source and target images at a resolution of 224×224 pixels. We then construct two KD trees [32] for efficient K nearest neighbor search. For each keypoint in the source image, we search for its corresponding K nearest neighbors on the target point map using the KD trees. In the semantic matching stage, we employ semantic feature maps of size 840×840 pixels. We upsample the K pixels identified during the geometric matching from the 224×224 resolution to the 840×840 resolution to obtain maximum cosine similarity between the semantic feature of the source keypoint and the features within the localized region of the target image for final correspondence.

Evaluation Datasets. We conduct experiments on widely used benchmarks SPair-71k [36], a challenging semantic correspondence dataset that contains a diverse set of 53,340 training pairs, 5,384 validation pairs, and 12,234 testing pairs spanning 18 object categories. This dataset presents significant variations in the viewpoint, scale, and object instances. We focus our evaluation on the **rigid body categories**, specifically, 11 categories in SPair-71k.

Evaluation Metric. We adopt the widely used percentage of correct keypoints **(PCK)** metric. The PCK measures the proportion of predicted keypoints that are correctly matched to the ground truth within a specified threshold.

A predicted keypoint is considered correct if it lies within a threshold of $\alpha \times \max(h, w)$ pixels from the ground truth keypoint, where $\alpha \in [0, 1]$ (e.g. $\alpha = 0.10$), and h and w represent height and width, respectively. Specifically, for the **SPair-71k** dataset, h and w are the dimensions of the *bounding box* of an instance (α_{bbox}) and we report all results using PCK *per point* for consistency in metric.

Model Variants and Baselines. we evaluate our baseline model and its variants, comparing them with state-of-the-art methods. Our baseline, **Ours (Geo Only)** only uses geometric matching for 1 nearest neighbor search. This baseline is enhanced with two variants on top of geometric matching for $K = 300$ nearest neighbors: **Ours (Geo + Semantic (DINOv2 ViT-S))** uses semantic matching from descriptors extracted by DINOv2 ViT-S and **Ours (Geo + Semantic (DINOv2 + SD)** uses semantic matching from descriptors extracted from DINOv2 ViT-B and Stable Diffusion (SD); these two descriptors are fused following [60].

We compare our methods with existing approaches: [50], [60], which only uses unsupervised semantic representation learned from the 2D image plane and [61] and [34], which incorporate geometric or spatial information into semantic representations. The methods we compare can therefore be grouped into 2 categories:

- Group **(a)**: Methods that rely on 2D representation
- Group **(b)**: Methods that inject geometric information.

[34, 61]. To clarify training and evaluation settings, for [60] and [61], we perform evaluations in its *zero-shot* setting. Except for Spherical Map [34] which uses the SPair-71k training split for training purposes without explicitly using

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Table 1. **Evaluation on SPair-71k per category:** The Per-Category and average Percentage of Correct Keypoints (PCK) at a threshold of 0.10 (PCK@0.1) on Spair-71k's test dataset are presented. Following [14, 37], we provide the PCK *per point* results on rigid-body categories in SPair-71k. The results with the highest PCK values are emphasized in **bold** font, whereas the second-highest values are underlined. $*$: Finetuned on SPair-71k training dataset.

Methods	Aero	Bike	Boat	Bottle	Bus	Car	Chair	Motor	Plant	Train	TV	**All**
DINOv2+NN [38, 60]	72.7	62.0	41.3	40.4	52.3	51.5	36.2	61.0	30.7	54.3	24.2	47.1
DIFT [50]	63.5	54.5	34.5	46.2	52.7	48.3	39.0	53.3	57.8	71.1	63.4	55.6
SD+DINO [60]	73.0	64.1	40.7	52.9	55.0	53.8	45.5	63.3	58.4	66.2	53.5	58.0
Spherical Maps* [34]	76.9	61.2	42.1	48.4	73.3	67.2	46.3	66.0	36.2	67.8	42.2	57.2
Spherical Maps + SD* [34]	74.8	64.5	45.6	52.7	**77.8**	71.4	47.7	67.6	49.9	78.5	59.1	63.9
Telling Left from Right [61]	78.0	**66.4**	44.5	60.1	66.6	60.8	53.2	66.1	58.9	**83.8**	55.5	64.9
Ours (Geo only)	84.3	54.0	43.9	59.8	49.1	83.5	**84.4**	72.5	65.3	79.1	75.0	69.6 (+4.7)
Ours (Geo + Semantic (DINOv2 ViT-S))	84.6	59.6	47.6	62.2	50.1	**84.4**	83.4	75.1	69.5	81.2	76.1	71.5 (+6.6)
Ours (Geo + Semantic (DINOv2 + SD))	**85.5**	62.2	**48.0**	**62.6**	50.3	84.3	84.0	**76.0**	**71.6**	81.7	**76.6**	**72.3** (+7.4)

Table 2. **Evaluation on SPair-71k at Different PCK Levels:** We present the performance on SPair-71k test split (rigid-body categories only) using PCK *per point* criteria at different thresholds. The highest PCK values are highlighted in **bold** font, and the second-highest values are underlined. $*$: Finetuned on SPair-71k training dataset.

	SPair-71k		
Methods	0.01	0.05	0.10
DINOv2+NN [38, 60]	5.0	32.1	47.1
DIFT [50]	9.5	45.2	55.6
SD+DINO [60]	7.2	43.0	58.2
Spherical Maps* [34]	5.8	39.4	57.8
Spherical Maps + SD* [34]	7.9	48.2	63.9
Telling Left from Right [61]	9.0	48.4	64.7
Ours (Geo only)	11.4	53.1	69.6
Ours (Geo + Semantic (DINOv2 ViT-S))	15.1	56.4	71.5
Ours (Geo + Semantic (DINOv2 + SD))	**19.1**	**59.1**	**72.3**

semantic correspondence labels, all other methods, including ours, do not use any data from the evaluation dataset.

4.2. Quantitative Results

Overall Semantic correspondence result. In Table 1, we present the result per category on PCK@0.1 on rigid body categories in SPair-71k. Our method achieves a gain of 14.3 and 7.4 compared to the best of the methods in group **(a)** and **(b)** respectively. Table 1 shows that without any additional semantic information, simply using 1 nearest neighbor matching, our baseline model outperforms the methods in group **(b)** by 4.7 and the methods in group **(a)** by 11.6.

In addition to Table 1, we also present the average PCK *per point* at thresholds of $[0.01, 0.05, 0.1]$ in Table 2. Under stricter metrics such as PCK@0.01 and PCK@0.05, our baseline method, without using any additional semantic information, outperforms the methods in group **(a)** by 1.9 and 7.9 percentage points, and in group **(b)** by 2.4 and 4.7 respectively. This improvement suggests that our deformation process not only accurately deforms to the correct shape, but also preserves semantic understanding. With an additional semantic feature, our method outperforms existing methods in PCK@0.01 and 0.05 by 9.6 and 10.7, which highlights the efficacy of our semantic matching module.

Semantic correspondence result in large viewpoint subset. We further evaluate the performance on a subset of SPair-71k of our choice, which contains image pairs with large or extreme viewpoint changes. This subset has 7 categories with 550 image pairs, and the result in Table 3 shows that our method is robust to extreme viewpoint changes and learns viewpoint-invariant correspondence by not exhibiting significant performance loss across all three of our model variants. In contrast, except for Spherical Maps and it's variant [34] which are trained on SPair-71k's training split, all other methods experience performance loss. Consequently, our method achieves a substantial improvement of 19.6 percentage points over other methods that have not used any data from SPair-71k.

Ablation study In Table 4, we report the performance differences in PCK at thresholds of $[0.01, 0.05, 0.1]$, as well as the improvements achieved using different semantic feature extractors. At the least strict threshold, PCK@0.1, we observe only minor improvement when changing the feature extractor from DINOv2 ViT-S to DINOv2 ViT-B + SD, at the cost of a 13x increase in runtime. However, the benefit of incorporating stronger semantic features becomes evident at stricter thresholds, PCK@0.05 and PCK@0.01, where using a more powerful semantic feature extractor yields approximately a 3 percent gain in performance. This shows that geometric matching already provides a solid prior in terms of promising performance in PCK@0.1 for stronger semantic features to capture finer-level semantic correspondences under stricter metrics.

4.3. Qualitative Results

We qualitatively compare our method with methods that rely on 2D pre-trained representation in Fig. 3, DIFT [50] and SD+DINO [61]. Showing that under extreme viewpoint variation, our method accurately finds the correspondence between image pairs.

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Table 3. **Evaluation on SPair-71k large viewpoint change subset per category:** We provide the *per point* PCK results on rigid-body categories with large viewpoint variation pairs in SPair-71k using PCK@0.1. The results with the highest PCK values are highlighted in **bold** font, and the second-highest values are underlined. $*$: Finetuned on SPair-71k training dataset.

Methods	Aero	Bike	Boat	Bus	Car	Chair	Motor	**All**
DINOv2+NN [38, 60]	67.2	62.9	39.4	20.4	22.2	21.9	60.6	51.1
DIFT [50]	53.4	48.3	5.6	4.8	1.8	8.7	35.0	31.9
SD+DINO [60]	66.5	61.9	20.3	21.0	17.6	13.8	59.9	47.1
Spherical Maps* [34]	77.2	4.5	38.6	**72.0**	53.7	40.8	71.0	63.2
Spherical Maps + SD* [34]	73.1	66.5	39.2	62.9	48.1	28.1	69.6	60.4
Telling Left from Right [61]	74.5	64.3	33.6	21.0	19.9	14.8	63.5	52.3
Ours (Geo only)	92.5	58.0	43.6	38.2	76.8	61.7	79.9	68.8
Ours (Geo + Semantic (DINOv2 ViT-S))	92.6	64.3	**48.1**	38.7	**78.2**	**62.8**	83.3	71.6
Ours (Geo + Semantic (DINOv2 + SD))	**93.6**	**66.8**	44.7	38.2	77.7	62.2	**83.3**	**71.9**

Table 4. **Ablation study on SPair-71k:** We report the PCK results for our baseline model and its variants as well as improvements in PCK at different threshold levels. Inference speed in fps on a single NVIDIA RTX 4090 GPU are also presented.

	SPair-71k			↑ SPair-71k			Latency
Model Variants	0.01	0.05	0.10	0.01	0.05	0.10	fps
Ours (Geo only)	11.4	53.1	69.6	-	-	-	19.9
Ours (Geo + Semantic (DINOv2 ViT-S))	15.1	56.4	71.5	3.7	3.3	1.9	5.4
Ours (Geo + Semantic (DINOv2 + SD))	19.1	59.1	72.3	4.0	2.7	0.8	0.4



Figure 3. Qualitative analysis between our method using geometry only and SD+DINO and DIFT

4.4. Limitations

Synthetic Data Quality and Ambiguity Despite promising results, our method has certain limitations. First, performance is heavily dependent on the quality of the 3D model. For instance, our lower performance on buses arises from overly simplistic ShapeNet meshes, leading to greater simulation-to-real gaps and ambiguity, particularly because the front and back appear similar. This ambiguity causes the network, under the $l2$ loss, to regress positions toward the midpoint of ambiguous ground-truth points.

Limitation in deformable categories Our current pipeline is limited to rigid-body objects for several reasons. First, most publicly available animal datasets provide only in-the-wild RGB images without accompanying depth maps or consistent camera poses in a unified coordinate system. Also, we assume orientation-aligned objects in 3D datasets. For nonrigid classes such as animals, where deformation introduces an effectively infinite number of degrees of freedom, defining a canonical 'orientation' is inherently difficult. One direction to address the non-rigid issue involves leveraging parametric models like SMAL[62] or SMPL[29]. These models can align real-world images to a common template, potentially offering a structured way to extend our method beyond rigid categories.

5. Conclusion

Semantic correspondence faces two main challenges: appearance variations and viewpoint changes. We address this challenge by extending DUSt3R for cross-instance viewpoint alignment. To address the lack of annotations and adapt DUSt3R for shape differences, we propose an efficient synthetic data annotation strategy using coordinate-aligned 3D models yet under various viewpoints. Our method introduces a novel hybrid pipeline that combines geometric alignment and feature-based local refinement. Despite training on synthetic data, it achieves superior zero-shot generalization to real-world scenarios, delivering significant performance improvements under drastic viewpoint changes while achieving high efficiency.

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